

Arun Krishnasamy

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EDUCATION

Monash University

Master of Business Analytics

Vaigai College of Engineering

Bachelors in Mechanical Engineering

Melbourne, VIC

Feb 2024 - Dec 2025

Madurai, India

Aug 2019 - Apr 2023

TECHNICAL PROJECTS

Melbourne Transit Policy Causal Impact Analysis | Independent Project (In Progress) Apr 2026 – Present

- Engineering a high-performance data pipeline to ingest, clean, and aggregate over **48 million rows** of Victorian Government TIRTL traffic sensor data, utilizing memory-optimized typing and Parquet compression.
- Conducting rigorous spatial and temporal EDA to validate sensor network reliability, engineering custom regressors to dynamically control for school holidays, public holidays, and directional commute flows.

Department of Treasury and Finance | Analyst Victorian Government

Jun 2025 - Oct 2025

- Designed a reproducible, mixed-frequency data ingestion pipeline in Python and R (targets), harmonizing **16,000+ time-series observations across 120+ regional indicators**.
- Implemented a robust 28-parameter schema registry to automate data transformation, missingness diagnostics, and quarterisation rules, demonstrating strong data governance and pipeline scalability.
- Prototyped a Bayesian VAR (MF-BVAR) model using PyMC and PyTensor, treating quarterly indicators as latent monthly states to improve the accuracy of real-time economic nowcasts.

Predicting the Probability of Death in an ICU (MIMIC-III)

May 2025

- Engineered an end-to-end predictive pipeline in R (tidymodels), processing over 650,000 diagnostic records across **14,500+** unique ICD-9 medical codes.
- Transformed high-dimensional categorical health data into numerical features using GloVe embeddings and comorbidity scores, successfully handling complex, sparse matrices at scale.
- Trained an ensemble machine learning model (Neural Networks, XGBoost, Random Forest) to predict patient mortality, achieving a **95% ROC-AUC** and ranking 7th on the public leaderboard.

PROFESSIONAL EXPERIENCE

Elegant Media

Business Analyst – Internship

Melbourne, VIC

Dec 2025 – Mar 2026

- Translated complex business workflows into detailed functional requirement documents and wireframes, bridging the gap between client needs and technical execution.
- Executed rigorous unit testing on mobile app builds, synthesizing technical findings into comprehensive test reports to ensure product quality prior to launch.

Woolworths Customer Fulfilment Centre

Online Team Member

Melbourne, VIC

July 2024 – Present

- Consistently exceed strict volume KPIs in a fast-paced environment, executing rapid-dispatch protocols to process 50+ product totes per session.
- Demonstrate strong adaptability and teamwork by dynamically reallocating across dispatch and produce departments to manage volume surges and ensure on-time delivery handovers.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Excel

Data Engineering & Wrangling: Data cleaning, transformation, preprocessing, and feature engineering

Machine & Deep Learning: Neural Networks, supervised/unsupervised learning, CNN, RNN, Bayesian inference, hypothesis testing

Libraries & Frameworks: pytorch, keras, pandas, numPy, sci-kit learn, tidymodels

Data Visualization: Tableau, Power BI, ggplot2, matplotlib, R shiny

Tools & Version Control: Git, GitHub, Vim

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate	2024
Google Data Analytics	2023